Cessna 441 Conquest II (C441)

2× Honeywell (Garrett) TPE331-8-401S/-402S or TPE331-10N (Dash 10 conversion), 636 shp flat-rated · 1–2 crew + up to 9 passengers seats · Typical 441 Conquest II (9,850 lb MTOW, 9,360 lb MLW) AFM procedures; TPE331 single-shaft engine with start locks, NTS and SRL. Condition levers are called speed levers in some manuals; autofeather is not fitted to all airframes. Dash 10 conversions share limits unless the STC supplement differs.

V-speeds & limits

Vmca	**91 KIAS**
Vr	**95–100 KIAS (weight dependent)**
V2	**approx 110 KIAS (108–112 by weight)**
Vxse	**109 KIAS**
Vyse	**118 KIAS (blue line; best single-engine rate of climb)**
Vy	**approx 130 KIAS; cruise climb 150–160 KIAS**
Vfe	**174 KIAS (APPROACH 15°) / 140 KIAS (LAND 30°)**
Vlo / Vle	**175 KIAS**
Va	**approx 170 KIAS (9,850 lb)**
Vmo / Mmo	**245 KIAS / 0.52 M**
Vs / Vso	**approx 85–90 / 75–80 KIAS (max weight)**
Vref	**95–105 KIAS (flaps LAND, weight dependent)**
Max differential	**6.3 psi; max operating altitude 35,000 ft**

Normal procedures

Before Start

Preflight inspection	**COMPLETE (prop start locks engaged – blades flat, not feathered)**
Passenger briefing	**COMPLETE (belts, cabin door / emergency exit, oxygen masks, no smoking)**
Seats, belts, shoulder harnesses	**ADJUST & LOCK**
Cabin door / emergency exit	**CLOSED & LOCKED (door light out)**
Parking brake	**SET**
Control lock	**REMOVED**
Landing gear switch	**DOWN**
Power levers	**GROUND IDLE**
Condition levers	**EMERGENCY SHUTOFF (aft)**
Avionics power	**OFF**
Bleed air switches	**OFF**
Ice protection switches	**OFF**
Firewall fuel shutoff switches	**OPEN (guards down, lights out)**
Fuel crossfeed	**OFF**
Battery switch	**ON (24 V minimum; external power as required)**
Inverter	**1 (or 2) – AC power check**
Annunciator panel	**PRESS TO TEST – all lights illuminate**
Fuel quantity	**CHECK both sides against plan (475 gal / approx 3,180 lb usable)**
Oxygen	**ON – pressure checked (approx 1,800 psi full), crew masks tested**
Cabin pressurisation	**CONTROLLER SET (cruise altitude, rate approx 500 fpm), dump switch NORMAL**
Beacon	**ON**

Engine Start

Propeller area	**CLEAR (both sides)**
Fuel boost pumps	**ON (L & R) – FUEL PRESS LOW lights out**
Power lever	**GROUND IDLE**
Condition lever	**LOW (forward of EMERGENCY SHUTOFF detent)**
Ignition	**NORMAL**
Start switch	**START (release – sequence is automatic)**
Fuel / ignition	**ON at approx 10% RPM (fuel flow indicated)**
Light-off	**EGT RISE within 10 sec – if none, abort (condition lever EMERGENCY SHUTOFF)**
EGT	**MONITOR – 770°C maximum during start**
NTS light	**ILLUMINATES during spool-up, then OUT (NTS check)**
Oil pressure	**RISING (40 psi minimum at idle)**
Starter	**CUT OUT approx 60% RPM (start light out)**
RPM	**STABILISED approx 65–75% (ground idle, LOW)**
Start locks	**RELEASE – power lever momentarily into REVERSE, then GROUND IDLE**
Propeller	**OFF LOCKS (blade angle / torque change noted)**
Generator	**ON – GEN OFF light out, charging**
Second engine	**REPEAT start sequence**
External power	**DISCONNECTED (if used)**
Engine instruments	**CHECK both (oil pressure 70–120 psi warm, oil temp 55–110°C, EGT, fuel flow)**
Avionics power	**ON**

Starter limits per AFM: allow the starter to cool between attempts; three consecutive battery starts maximum. A start with a feathered propeller (locks not engaged) is a hot, high-load start – use the unfeather pump to bring the blades flat first, or follow the AFM feathered-start procedure.

Before Taxi

Bleed air switches	**ON (both) – cabin temperature as required**
Inverter	**CHECK 1 and 2 (select each, AC instruments powered)**
Flight instruments	**CHECK & SET (altimeters, HSI, attitude indicators, standby instruments)**
Avionics / GPS / radios	**SET**
Electric pitch trim	**CHECK (each half of split switch alone – no motion; together – motion; disconnect stops trim)**
Autopilot / yaw damper	**TEST, ENGAGE, OVERPOWER & DISCONNECT – then OFF**
Trims (pitch, rudder, aileron)	**SET for takeoff**
Condition levers	**LOW (taxi RPM approx 65–75%)**
Brakes	**CHECK on initial roll**
Power levers	**GROUND IDLE / BETA as required – REVERSE only at taxi speeds**

Before Takeoff

Parking brake	**SET**
Flight controls	**FREE & CORRECT**
Condition levers	**HIGH – 100% RPM both (check within 1%)**
SRL computers	**CHECK – switch OFF: SRL OFF light ON, EGT indication decreases; switch ON: light OUT**
Overspeed governor / NTS	**CHECK per AFM (or NTS lights checked during start)**
Autofeather (if installed)	**TEST, then ARM**
Engine instruments	**CHECK both (oil 70–120 psi, EGT stable, fuel flow matched, torque matched)**
Generators	**CHECK both charging, voltage 28 V, load balanced**
Ice protection	**CHECK (prop deice ammeter cycles, surface deice one cycle, windshield heat, pitot / stall heat) – then AS REQUIRED**
Engine inlet anti-ice	**ON if OAT below 5°C in visible moisture (expect EGT rise)**
Ignition	**NORMAL (CONTINUOUS for standing water, heavy rain or icing)**
Fuel boost pumps	**ON both; crossfeed OFF; FUEL PRESS LOW lights out**
Firewall fuel shutoffs	**OPEN**
Bleed air	**ON both; cabin pressurisation controller SET**
Wing flaps	**UP or APPROACH (15°) per performance**
Trims	**SET takeoff**
Landing gear switch	**DOWN – three green**
Pitot / static heat	**ON**
Lights	**AS REQUIRED**
Transponder	**ALT**
Cabin door / exit	**LOCKED – lights out**
Takeoff briefing	**COMPLETE (Vmca 91 / Vr 95–100 / V2 110 / Vyse 118; abort before rotation; after liftoff gear up, confirm NTS / feather)**
Condition levers	**HIGH (100%) for takeoff**
Power levers	**SET TAKEOFF TORQUE – do not exceed 100% torque or 770°C EGT**

After Takeoff / Climb

Landing gear	**UP (positive rate, below 175 KIAS) – lights out**
Wing flaps	**UP (above V2 / 110 KIAS)**
Climb power	**SET – torque within limits, EGT 770°C max, 100% RPM**
Airspeed	**130 KIAS (best rate) / 150–160 KIAS cruise climb**
Autofeather (if installed)	**OFF above safe altitude per AFM**
Yaw damper / autopilot	**AS REQUIRED (above minimum engagement altitude)**
Cabin pressurisation	**CHECK climbing approx 500 fpm, differential increasing – 6.3 psi max**
Engine instruments	**CHECK both**
Ice protection	**AS REQUIRED**
Altimeters	**SET standard at transition**

Cruise

Condition levers	**96–100% RPM as desired (96% typical cruise)**
Cruise power	**SET per AFM table – torque / EGT (770°C max, 650–700°C typical) / fuel flow**
Propellers	**SYNCHRONISE (synchrophaser ON if fitted)**
Fuel quantity / balance	**CHECK both sides against plan – crossfeed to balance as required**
Cabin pressurisation	**CHECK cabin altitude / differential (6.3 psi max), rate zero**
Oxygen	**PRESSURE CHECK (above FL250 crew mask within reach / per regulations)**
Engine instruments	**CHECK both (oil pressure / temperature, EGT, torque, RPM)**
Electrical	**CHECK generator loads balanced, inverter**
Ice protection	**AS REQUIRED (engine anti-ice ON in visible moisture below 5°C)**

Descent

Altimeters	**SET**
Cabin pressurisation	**SET landing field elevation, rate approx 500 fpm**
Power	**REDUCE as required – power levers never below FLIGHT IDLE in flight**
Airspeed	**BELOW 245 KIAS / 0.52 M; plan below 175 KIAS for gear**
Fuel crossfeed	**OFF (balanced)**
Ice protection	**AS REQUIRED (engine anti-ice ON in visible moisture below 5°C)**
Approach briefing	**COMPLETE (incl. single-engine go-around decision point)**
Seat belts / harnesses	**SECURE – passengers briefed**

Before Landing

Fuel boost pumps	**ON both; crossfeed OFF**
Condition levers	**HIGH (100% RPM)**
Ignition	**CONTINUOUS if standing water / heavy precipitation**
Airspeed	**BELOW 175 KIAS**
Landing gear	**DOWN – three green, hydraulic pressure normal**
Wing flaps	**APPROACH (15°) below 174 KIAS; LAND (30°) below 140 KIAS**
Cabin pressurisation	**CHECK differential near zero before touchdown**
Autopilot / yaw damper	**OFF before landing**
Airspeed	**Vref 95–105 KIAS (flaps LAND)**
Landing lights	**ON**
Landing gear	**THREE GREEN – confirmed on final**

After Landing

Power levers	**GROUND IDLE (lift over flight-idle gate) – REVERSE as required, idle below taxi speed**
Condition levers	**LOW**
Wing flaps	**UP**
Ice protection / pitot heat	**OFF**
Ignition	**NORMAL**
Landing / strobe lights	**AS REQUIRED**
Transponder	**STBY / as required**
Trims	**SET takeoff**

Shutdown

Parking brake	**SET**
Power levers	**GROUND IDLE**
Condition levers	**LOW – 1 minute cool-down, EGT stabilised**
Avionics power	**OFF**
Bleed air / environmental	**OFF**
Ice protection / lights	**OFF**
Condition levers	**EMERGENCY SHUTOFF (fuel cutoff)**
Power levers	**REVERSE as RPM decays – hold until propellers stop (engages start locks)**
NTS lights	**ILLUMINATE during coast-down (NTS check)**
Propellers	**STOPPED on start locks – blades flat, not feathered**
Power levers	**GROUND IDLE**
Fuel boost pumps	**OFF**
Generators / inverter	**OFF**
Battery switch	**OFF**
Control lock	**INSTALL**
Chocks / covers	**AS REQUIRED**

If the start locks did not engage the propeller feathers as it stops. Note it for the next start (unfeather pump or AFM feathered-start procedure).

Abnormal procedures

Generator Failure (GEN OFF light)

L or R GEN OFF annunciator, loadmeter zero

Affected generator switch ..... **RESET, then ON**

▶ Generator restored?

Yes:

Loadmeters / voltmeter ..... **MONITOR both**

No – one generator failed:

Affected generator switch ..... **OFF**

Electrical load ..... **REDUCE to within one generator's capacity (ice protection, air conditioning)**

Deferred: One generator inoperative – electrical load reduced, land as soon as practical. ..... **NOTED**

No – both failed:

Both generator switches ..... **OFF**

Non-essential electrical ..... **OFF – battery only, approx 30 min endurance**

Inverter ..... **ONE only**

Landing gear ..... **PLAN early extension while battery power remains (hand pump if needed)**

Deferred: Both generators failed – battery only; extend gear early, flaps may be unavailable, land as soon as possible. ..... **NOTED**

Inverter Failure (INVERTER / AC light)

Inverter select switch ..... **OTHER INVERTER (1 to 2 or 2 to 1)**

AC-powered instruments ..... **CHECK restored (attitude / HSI / flight director as applicable)**

▶ AC power restored?

Yes:

Deferred: One inverter failed – no AC backup; land as soon as practical. ..... **NOTED**

No:

Standby attitude indicator / magnetic compass ..... **USE – cross-check with GPS**

Autopilot ..... **OFF if coupled to failed instruments**

IMC ..... **EXIT / AVOID**

Deferred: Both inverters failed – partial panel, no AC instruments; prefer VMC approach. ..... **NOTED**

Low Oil Pressure (one engine)

OIL PRESS LOW light or pressure below 70 psi above idle (40 psi minimum)

Oil pressure / oil temperature ..... **CHECK and cross-check the affected engine**

▶ Pressure below 40 psi, falling, or oil temperature rising?

Yes – genuine:

Affected engine power ..... **REDUCE to minimum required**

Engine ..... **SHUT DOWN & FEATHER before seizure if altitude and terrain allow (Engine Failure in Flight)**

A seizing TPE331 may not feather – shut down early while oil pressure remains.

Deferred: Low oil pressure – precautionary shutdown, single-engine landing, land as soon as possible. ..... **NOTED**

No – indication suspect:

Flight ..... **CONTINUE, monitor pressure and temperature**

Deferred: Oil pressure indication suspect – maintenance check required. ..... **NOTED**

CHIP DETECTOR light

Affected engine oil pressure / temperature ..... **MONITOR closely**

Affected engine power ..... **REDUCE to minimum practical**

▶ Other abnormal indications (vibration, oil pressure, temperature)?

No:

Deferred: Chip detector – land as soon as practical, maintenance inspection required. ..... **NOTED**

Yes:

Engine ..... **SHUT DOWN & FEATHER if altitude and terrain allow**

Deferred: Chip detector with secondary indications – precautionary shutdown, land as soon as possible. ..... **NOTED**

FUEL PRESS LOW / Crossfeed

L or R FUEL PRESS LOW light, or fuel imbalance

Affected side fuel boost pump ..... **CHECK ON (switch / breaker)**

Fuel quantity ..... **CHECK both sides**

▶ Light extinguished?

Yes:

Deferred: Fuel boost pump cycled – pump suspect, maintenance check. ..... **NOTED**

No – pump failed, fuel available:

Fuel crossfeed ..... **ON – engine fed from other side's pump**

Altitude ..... **LIMIT per AFM (engine-driven pump suction feed) if crossfeed not used**

Deferred: Fuel boost pump failed – crossfeed in use, monitor balance. ..... **NOTED**

No – fuel low on that side:

Fuel crossfeed ..... **ON to balance**

Fuel endurance ..... **RECOMPUTE – divert as required**

Deferred: Fuel imbalance / low – crossfeed used, land with crossfeed OFF where possible. ..... **NOTED**

Crossfeed is for balancing and single-engine operation: do not take off or land with crossfeed ON unless required.

NTS / Autofeather Inoperative Indication

NTS light fails to illuminate on start or shutdown, or NTS light ON in flight

▶ When?

On the ground – check failed:

Engine ..... **SHUT DOWN – do not fly without a satisfactory NTS check**

Without NTS an engine failure leaves a nearly flat-pitch windmilling propeller with very high drag until manually feathered.

In flight – NTS light ON:

Affected engine torque / EGT / RPM ..... **CHECK – NTS light means negative torque (engine not producing power)**

Affected engine ..... **IF FAILED: condition lever EMERGENCY SHUTOFF (Engine Failure in Flight)**

If engine normal ..... **POWER LEVER ADVANCE slightly (light may show at flight idle in descent)**

Deferred: NTS light in flight – if engine normal, system check required after landing. ..... **NOTED**

In flight – autofeather inoperative light:

Autofeather switch ..... **OFF then ARM (reset once)**

Engine failure plan ..... **MANUAL FEATHER with condition lever – brief for approach**

Deferred: Autofeather inoperative – manual feather required on engine failure. ..... **NOTED**

Bleed Air / Pressurisation Fault

BLEED AIR FAIL / overheat light, cabin altitude climbing or differential high

▶ Indication?

Bleed air fail / overheat light:

Affected bleed air switch ..... **OFF**

Pressurisation ..... **CHECK maintained on remaining bleed source**

Deferred: One bleed source off – reduced pressurisation / heating capacity, maintenance check. ..... **NOTED**

Cabin not holding – altitude climbing:

Bleed air switches ..... **CHECK both ON**

Pressurisation controller ..... **MANUAL – control outflow valve to hold cabin**

Cabin dump switch ..... **CHECK NORMAL**

Descent ..... **INITIATE to keep cabin altitude below 10,000 ft; oxygen if required**

Deferred: Pressurisation on manual / limited – altitude restricted. ..... **NOTED**

Differential above 6.3 psi:

Pressurisation controller ..... **MANUAL – open outflow valve to reduce differential**

Bleed air ..... **ONE SOURCE OFF if differential still high**

Descent ..... **INITIATE**

Deferred: Over-pressurisation event – structural inspection required. ..... **NOTED**

SRL OFF light / SRL Failure in Flight

SRL OFF annunciator on, or EGT indication erratic

SRL switch (affected engine) ..... **OFF then ON (reset once)**

▶ SRL OFF light out and EGT indication normal?

Yes:

EGT ..... **MONITOR – compare with other engine**

No – SRL OFF light stays on:

SRL switch ..... **OFF**

EGT limit ..... **REDUCED – use the SRL-OFF EGT chart / placard (indicated EGT is uncompensated; approx 650°C typical)**

Power ..... **SET by torque, respecting the reduced EGT limit – expect less power available at altitude**

Deferred: SRL inoperative – reduced EGT limit, reduced power available including for a go-around. ..... **NOTED**

No – EGT erratic without light:

SRL switch ..... **OFF – SRL OFF light ON**

EGT limit ..... **REDUCED per SRL-OFF chart / placard**

Torque / fuel flow ..... **CROSS-CHECK with other engine for power setting**

Deferred: SRL switched off for erratic EGT – reduced EGT limit, maintenance required. ..... **NOTED**

Icing

Ice accretion or visible moisture below 5°C

Engine inlet anti-ice ..... **ON both (EGT rise expected – adjust power)**

Ignition ..... **CONTINUOUS**

Pitot / static / stall warning heat ..... **ON**

Propeller deice ..... **ON – ammeter cycling**

Windshield anti-ice ..... **ON**

Surface deice ..... **CYCLE when approx 1/2 in ice has formed**

Autopilot ..... **OFF periodically to feel control forces**

Exit icing ..... **CLIMB / DESCEND / TURN**

▶ Residual ice on airframe?

No:

Ice protection ..... **AS REQUIRED**

Yes:

Approach ..... **FLAPS APPROACH (15°) maximum, Vref + 10–15 KIAS, longer runway**

Stall warning ..... **UNRELIABLE – avoid steep turns and low speed**

Deferred: Residual airframe ice – 15° flap landing, increased approach speed and landing distance. ..... **NOTED**

Pitot Heat Failure

PITOT HEAT light or airspeed disagreement in icing

Pitot heat switch / circuit breaker ..... **CHECK ON / RESET once**

Airspeed indications ..... **CROSS-CHECK left / right and GPS groundspeed**

▶ Airspeed reliable?

Yes:

Icing conditions ..... **AVOID**

Deferred: Pitot heat inoperative one side – avoid icing, cross-check airspeed. ..... **NOTED**

No:

Autopilot / flight director ..... **OFF if driven by failed air data**

Pitch / power ..... **FLY KNOWN SETTINGS – use other side airspeed, GPS, attitude**

Exit icing ..... **IMMEDIATELY**

Deferred: Airspeed unreliable – fly known pitch / power settings, Vref + 10 KIAS on the good side. ..... **NOTED**

Hydraulic Pressure Low

HYD PRESS LOW light or pressure below normal

Hydraulic pump circuit breaker ..... **CHECK / RESET once**

Landing gear ..... **LEAVE in present position**

▶ Pressure restored?

Yes:

Hydraulic pressure ..... **MONITOR**

Deferred: Hydraulic pressure fluctuation – maintenance check. ..... **NOTED**

No:

Gear extension ..... **PLAN Emergency Gear Extension (hand pump) early on approach**

Brakes ..... **NORMAL (independent system) – check pedal pressure**

Deferred: Hydraulic pressure lost – emergency gear extension, do not retract after extension. ..... **NOTED**

Gear Unsafe / In-Transit Light

Red gear light remains on after retraction or extension

Airspeed ..... **BELOW 175 KIAS**

Gear switch / circuit breaker ..... **CHECK position / RESET once**

Landing gear ..... **CYCLE once to the desired position**

▶ Gear indication?

Three green – unsafe light out:

Deferred: Gear fault indication – do not retract after landing; maintenance inspection required. ..... **NOTED**

Will not retract – down and locked:

Landing gear switch ..... **DOWN – leave down**

Airspeed ..... **BELOW 175 KIAS**

Deferred: Gear will not retract – gear down, reduced range and single-engine performance, land as soon as practical. ..... **NOTED**

Will not indicate down:

Gear indicator bulbs ..... **CHECK (press to test / swap bulb if possible)**

Emergency gear extension ..... **PERFORM (hand pump)**

Tower ..... **REQUEST visual check if available**

Deferred: Gear position unconfirmed – emergency extension performed, soft touchdown on longest runway. ..... **NOTED**

Emergency procedures

Engine Failure / Abort During Takeoff

Before rotation, or airborne with runway remaining

Power levers (memory) ..... **GROUND IDLE then REVERSE (both)**

Brakes (memory) ..... **MAXIMUM – stop straight ahead**

▶ Airborne?

No – on the runway:

Condition levers ..... **EMERGENCY SHUTOFF (both) once stopped or if fire**

Firewall fuel shutoffs ..... **CLOSED**

Fuel boost pumps ..... **OFF**

Battery / generators ..... **OFF**

Yes – runway remaining:

Land (memory) ..... **STRAIGHT AHEAD, gear down**

Power levers ..... **GROUND IDLE / REVERSE after touchdown**

Condition levers / firewall shutoffs ..... **EMERGENCY SHUTOFF / CLOSED after stop**

Yes – no runway remaining:

Go to Engine Failure After Liftoff.

Below Vmca (91 KIAS) directional control cannot be assured with takeoff power on one engine. Reverse on one engine only produces a strong yaw – use symmetric reverse or none.

Engine Failure After Liftoff

Airborne, insufficient runway to land ahead

LAND AS SOON AS POSSIBLE

Power levers (memory) ..... **MAXIMUM TORQUE (do not exceed 770°C EGT)**

Condition levers (memory) ..... **HIGH (100%)**

Landing gear (memory) ..... **UP**

Airspeed (memory) ..... **V2 110 KIAS to clear obstacles, then Vyse 118 KIAS – never below 91 KIAS**

NTS / autofeather (memory) ..... **CONFIRM – inoperative propeller moving toward feather (yaw controllable)**

Inoperative engine (memory) ..... **IDENTIFY – dead foot, dead engine**

Inoperative engine power lever (memory) ..... **VERIFY – retard, no change**

Inoperative engine condition lever (memory) ..... **EMERGENCY SHUTOFF (feather / fuel cutoff)**

Wing flaps ..... **UP (above 110 KIAS)**

Bank ..... **3–5° toward operating engine, ball half out**

Trim ..... **RUDDER & AILERON for straight flight**

Climb ..... **118 KIAS to safe altitude, then Engine Failure in Flight (secure items)**

NTS only reduces drag – it does not fully feather the propeller. Feather with the condition lever once the failed engine is confirmed.

Deferred: One engine inoperative after takeoff – single-engine approach and landing, no go-around once committed. ..... **NOTED**

Engine Failure in Flight

Loss of power on one engine at altitude

LAND AS SOON AS POSSIBLE

Power levers (memory)	**AS REQUIRED (within torque / EGT limits)**
Condition levers (memory)	**HIGH (100%)**
Airspeed (memory)	**118 KIAS (Vyse) minimum; gear and flaps UP**
Inoperative engine (memory)	**IDENTIFY – dead foot, dead engine**
Inoperative engine power lever (memory)	**VERIFY – retard, no change**
Inoperative engine condition lever (memory)	**EMERGENCY SHUTOFF (feather / fuel cutoff)**
Inoperative engine firewall fuel shutoff	**CLOSED**
Inoperative engine fuel boost pump	**OFF**
Inoperative engine generator	**OFF**
Inoperative engine bleed air	**OFF**
Inoperative engine ice protection	**OFF (prop deice, inlet anti-ice)**
Propeller synchrophaser	**OFF (if fitted)**
Electrical load	**REDUCE (single generator)**
Fuel crossfeed	**AS REQUIRED to use fuel from the dead-engine side**
Bank / rudder / trim	**3–5° toward operating engine, ball half out, trimmed**
Cabin pressurisation	**CHECK – single bleed source, differential may reduce**

▶ Cause known (fuel / inadvertent shutoff) and within air-start envelope?

Yes – attempt air start:

Use the Air Start (Starter Assisted) procedure. If unsuccessful, leave the engine secured.

No – leave secured:

ATC	**ADVISE – nearest suitable airport, drift-down altitude checked**

Deferred: One engine inoperative – single-engine approach and landing, single generator / bleed source, no go-around once committed. **NOTED**

Engine Fire in Flight

FIRE warning light or visible fire

LAND IMMEDIATELY

Affected engine condition lever (memory)	**EMERGENCY SHUTOFF (feather / fuel cutoff)**
Affected engine firewall fuel shutoff (memory)	**CLOSED**
Affected engine fuel boost pump (memory)	**OFF**
Affected engine bleed air (memory)	**OFF**
Fire extinguisher (if installed) (memory)	**DISCHARGE to affected engine**
Affected engine generator	**OFF**
Affected engine power lever	**FLIGHT IDLE**
Affected engine ice protection	**OFF**
Airspeed	**118 KIAS or higher – increase if fire persists**
Cabin pressurisation / air	**CHECK for smoke – dump / descend as required**
Landing	**NEAREST SUITABLE AIRPORT – do not attempt restart**

Deferred: Engine fire – engine secured, single-engine landing, fire services requested. **NOTED**

Engine Fire During Start / On the Ground

Condition lever (memory)	**EMERGENCY SHUTOFF**
Firewall fuel shutoff (memory)	**CLOSED**
Starter (memory)	**CONTINUE CRANKING to purge fuel / lower EGT (if engine turning)**
Fuel boost pumps	**OFF**
Fire extinguisher (if installed)	**DISCHARGE**
Other engine	**SHUT DOWN (condition lever EMERGENCY SHUTOFF)**
Battery / generators	**OFF**
Aircraft	**EVACUATE – portable extinguisher on the fire from upwind**

Do not attempt another start after a fire until the engine has been inspected.

Air Start (Starter Assisted)

Restart attempt with cause identified; within envelope (typically below 20,000 ft, 120–150 KIAS)

The TPE331 is a single-shaft engine: a feathered propeller will not windmill it to starting RPM. Every air start is starter assisted. No unassisted windmilling restart exists.

Airspeed / altitude ..... **WITHIN AIR-START ENVELOPE (approx 120–150 KIAS, below 20,000 ft)**

Firewall fuel shutoff ..... **OPEN**

Fuel boost pump ..... **ON**

Generator (affected engine) ..... **OFF**

Electrical load ..... **REDUCE (starter draws from battery / other generator)**

Power lever ..... **FLIGHT IDLE**

Condition lever ..... **LOW (out of EMERGENCY SHUTOFF)**

Unfeather pump ..... **ON until propeller leaves feather / RPM rising**

Start switch ..... **START**

Ignition / fuel ..... **ON automatically at approx 10% RPM**

EGT ..... **MONITOR – 770°C maximum; abort if no light-off within 10 sec or EGT rising through limit**

Starter ..... **CUTS OUT approx 60% RPM – unfeather pump OFF**

▶ Engine running normally?

Yes:

- Condition lever ..... **HIGH (match RPM) after oil pressure / temperature normal**
- Power lever ..... **ADVANCE slowly – match torque**
- Generator / bleed air / ice protection ..... **ON as required**
- Fuel crossfeed ..... **AS REQUIRED to rebalance**
- Deferred: Engine restarted in flight – land as soon as practical, cause investigation required. ..... **NOTED**

No:

- Condition lever ..... **EMERGENCY SHUTOFF**
- Start switch / unfeather pump ..... **OFF**
- Firewall fuel shutoff / boost pump ..... **CLOSED / OFF**

Leave the engine secured – complete Engine Failure in Flight. Allow starter cooling before any further attempt.

Emergency Descent

Power levers (memory) ..... **FLIGHT IDLE**

Condition levers (memory) ..... **HIGH (100%)**

Landing gear (memory) ..... **DOWN (below 175 KIAS)**

Wing flaps (memory) ..... **UP**

Airspeed (memory) ..... **175 KIAS gear down; up to 245 KIAS / 0.52 M clean if structure not in doubt**

Autopilot ..... **OFF**

Oxygen masks ..... **ON if cabin altitude above 10,000 ft**

ATC ..... **ADVISE**

Level-off ..... **10,000 ft or MEA – power as required**

Rapid Decompression / CABIN ALT Warning

CABIN ALT light, cabin altitude above 10,000 ft or rapid differential loss

LAND AS SOON AS PRACTICAL

Crew oxygen masks (memory) ..... **ON – 100%**

Microphone (memory) ..... **OXYGEN MASK**

Passenger oxygen (memory) ..... **ON / masks deployed – passengers don masks**

Emergency descent (memory) ..... **INITIATE to 10,000 ft or MEA**

Cabin pressurisation ..... **CHECK controller mode, dump switch NORMAL, bleed air switches ON**

▶ Cabin pressure recoverable (controller / bleed fault)?

Yes:

- Pressurisation ..... **MANUAL mode / other bleed source as required**
- Altitude ..... **LIMIT to keep cabin altitude below 10,000 ft**
- Deferred: Pressurisation fault – fly unpressurised or on manual control, oxygen used; maintenance required. ..... **NOTED**

No – structural / door:

- Altitude ..... **10,000 ft or MEA; airspeed reduced if structural damage suspected**
- Oxygen ..... **MONITOR supply duration**
- Deferred: Unpressurised with possible structural damage – land as soon as possible, reduced airspeed. ..... **NOTED**

Smoke / Electrical Fire

LAND AS SOON AS POSSIBLE

Crew oxygen masks (memory) ..... **ON – 100%, microphone to mask**

Avionics power (memory) ..... **OFF**

Non-essential electrical (memory) ..... **OFF**

▶ Smoke source?

Electrical:

Fire extinguisher ..... **USE on source**

Circuit breakers ..... **IDENTIFY & PULL affected circuit**

▶ Fire out?

Yes:

Essential electrical / avionics ..... **ON one at a time, watching for smoke**

Deferred: Electrical fire – suspect circuits isolated, affected systems unavailable. ..... **NOTED**

No:

Battery & generators ..... **OFF**

Inverter ..... **OFF**

Standby instruments only. Gear by emergency hand pump, flaps inoperative – plan a flaps-up landing with no landing lights.

Deferred: Electrical power off – emergency gear extension, flaps-up landing, no radios / lights. ..... **NOTED**

Environmental / bleed air:

Bleed air switches ..... **OFF one at a time to identify source**

Suspect bleed air ..... **OFF – pressurise on remaining engine**

Deferred: Bleed air smoke – one bleed source off, reduced pressurisation / heating. ..... **NOTED**

Smoke removal ..... **DESCEND below 10,000 ft, pressurisation DUMP / cabin air open as required**

Emergency Gear Extension

Gear does not extend hydraulically / HYD PRESS LOW

Airspeed ..... **BELOW 175 KIAS – reduce to approx 140 KIAS**

Landing gear circuit breaker / hydraulic pump ..... **PULL / OFF (per AFM)**

Landing gear switch ..... **DOWN**

Emergency gear extension valve / release ..... **EMERGENCY (if fitted – gear free-falls partially)**

Emergency hand pump ..... **PUMP until three green and pressure felt (handle in pedestal / floor)**

Landing gear indicator ..... **THREE GREEN**

Hand pump handle ..... **STOW**

Do not reset the hydraulic pump breaker or move the gear switch after an emergency extension. Leave the gear down.

▶ Three green obtained?

Yes:

Deferred: Gear extended by emergency means – do not retract; gear / hydraulic inspection required. ..... **NOTED**

No:

Runway ..... **LONGEST available, emergency services advised**

Landing ..... **FLAPS LAND, minimum speed, condition levers EMERGENCY SHUTOFF and firewall shutoffs CLOSED just before touchdown, battery OFF**

Deferred: Gear-up / partial gear landing – longest runway, engines shut down on short final, evacuate after stopping. ..... **NOTED**

Single-Engine Approach, Landing & Go-Around

One engine secured

Operating engine fuel boost pump ..... **ON; crossfeed as required, then OFF for landing**

Operating engine condition lever ..... **HIGH (100%)**

Operating engine bleed air ..... **ON (pressurisation on single source)**

Airspeed ..... **118 KIAS (Vyse) until landing assured**

Approach ..... **NORMAL glidepath or slightly high – avoid low, flat approaches**

Landing gear ..... **DOWN when landing assured – three green**

Wing flaps ..... **APPROACH (15°) when landing assured; LAND (30°) on short final only**

Airspeed ..... **REDUCE below 118 KIAS only on short final – Vref 95–105 KIAS**

Power lever ..... **REDUCE smoothly – anticipate yaw change with power**

▶ Go-around required?

No:

- Landing ..... **NORMAL – reverse NOT recommended (asymmetric); brakes as required**
- After stop ..... **Clear runway on one engine only if controllable, otherwise shut down and await tow**

Yes – above decision height:

- Operating engine power lever ..... **MAXIMUM TORQUE (770°C EGT max)**
- Wing flaps ..... **APPROACH (15°) then UP above 110 KIAS**
- Landing gear ..... **UP (positive climb)**
- Airspeed ..... **118 KIAS (Vyse)**
- Bank ..... **3–5° toward operating engine**

A single-engine go-around below approx 200 ft AGL, or once LAND flaps are extended, should not be attempted.

Propeller Overspeed / Uncontrolled RPM

RPM above 101% and not responding to the condition lever

LAND AS SOON AS POSSIBLE

Affected engine power lever (memory) ..... **REDUCE**

Affected engine condition lever (memory) ..... **LOW**

Airspeed (memory) ..... **REDUCE (lower pitch attitude load on the propeller)**

Oil pressure (affected engine) ..... **CHECK – governor is oil operated**

▶ RPM controllable at or below 100%?

Yes:

- Power ..... **SET minimum that keeps RPM in limits; condition lever as required**
- Propeller synchrophaser ..... **OFF**
- Deferred: Propeller governor fault – RPM controlled by power lever and airspeed, land as soon as possible. ..... **NOTED**

No – RPM above 105% / uncontrolled:

- Affected engine condition lever ..... **EMERGENCY SHUTOFF (feather)**
- Engine ..... **SECURE per Engine Failure in Flight**

Expect high drag if the propeller does not feather. Maintain 118 KIAS or above.

Deferred: Engine shut down for propeller overspeed – single-engine landing, possible unfeathered propeller drag. ..... **NOTED**